Xihang Shan

Xiamen University | 19020232202354@stu.xmu.edu.cn | github.com/AndyShan11

Research focus: machine learning, algorithms, artificial intelligence, AI agents, and mathematical modeling.

Education

Xiamen University, School of Mathematical Sciences Sep 2023 – Expected Jun 2027
B.S. in Mathematics and Applied Mathematics | **GPA: 86.08/100** **Rank: 7/39**

Selected Manuscripts

1. **PRCD-MAP: Learning How Much to Trust Imperfect Priors in Causal Discovery** [PDF]
   Xihang Shan, Da Zhou* *ICLR 2027, under review; NeurIPS 2026 rejected (reviews: 4/4/5).*
   Field / problem: Causal discovery: expert and LLM edge priors are useful but can be wrong, with reliability varying across edges.
   Innovation: Empirical-Bayes calibration and propagation along the prior graph assign each edge a trust weight that adjusts the prior penalties in structure learning. Data-supported edges can guide discovery; conflicting edges are down-weighted, and uninformative priors revert toward the data-only model.
2. **DELTA-NAR: Learning What to Reuse and What to Recompute in Neural Algorithmic Reasoning**
   Xihang Shan, Ye Luo* [PDF] *AAAI 2027, under review (review scores: 5/6).*
   Field / problem: Neural algorithmic reasoning: small graph edits leave much of a previous solution valid, yet standard models recompute it.
   Innovation: CLRS-Delta makes the graph edit and previous solution explicit inputs. An affected-region gate locates what changed, while a copy-safety gate reuses valid state and repairs stale paths; this avoids both full recomputation and blindly copying an invalid old answer.

*Corresponding author.

Hackathon Project

CourtVision – AI Basketball Tactical Analysis [Live Demo] [Competition] Oct 2026
NBA × AWS Broadcast Code 12 Hackathon, Team 18. Built an AI agent that turns basketball video and court data into evidence-linked tactical stories.

- **Analysis and presentation:** Identify candidate key possessions, add tactical-board overlays and shot summaries, and generate commentary synchronized with the original footage; unknown player identities or metrics remain explicitly unclaimed.
- **Cloud pipeline:** Built an MP4 upload-to-render workflow using Amazon S3, Bedrock AgentCore, AWS Batch, Lambda, DynamoDB, and CloudFront, producing a reviewable and downloadable annotated video.

Research Experience

- **Summer Research – LocSource**
  Hong Kong University of Science and Technology (HKUST) Jul 2026
  Worked with Prof. Can Yang. **Problem:** Cell segmentation can assign individual RNAs to the wrong cell.
  - **Innovation:** Compare the assigned cell with nearby donor candidates using expression evidence that excludes the RNA being judged, preventing self-confirmation.
  - **Problem solved:** Aggregate evidence across a cell interface before proposing a change; uncertain cases abstain, and every RNA-level proposal is reversible without overwriting its original assignment.
- **National Undergraduate Innovation and Entrepreneurship Training Program**
  Xiamen University | **National-level project** Nov 2025 – Present
  Field / problem: Causal discovery with unreliable expert or LLM priors. With Prof. Da Zhou, built controlled prior-quality tests to check when edge-level trust should use a prior and when it should fall back to data-only discovery.
- **Research Collaboration – Graph Reasoning and Neural Algorithms**
  Xiamen University Jul 2024 – Present
  Field / problem: Neural graph algorithms after local edits. With Prof. Ye Luo, made the previous solution a model input and separated safe reuse from local repair, addressing unnecessary recomputation and stale-path copying.
- **Summer School Project – Structured Methods for Boolean Functions**
  Institute of Information Engineering, Chinese Academy of Sciences Jul 2025
  Field / problem: Boolean-function structure hidden by exponential search. Recast affine approximation and covering as checkable spectral/derivative conditions, narrowing search to candidates with explicit certificates.
  Outstanding Student, top 10 of 60 participants.

Honors

- Fujian Province First Prize, National Undergraduate Mathematical Modeling Contest, Team Leader, 2025.
- Fujian Province Third Prize, National Undergraduate Mathematical Modeling Contest, Team Leader, 2024.

Technical Skills

Programming	Python, C++17, LaTeX, OpenMP
ML/Graphs	PyTorch, PyG, DGL, vLLM; causal discovery, bandits, knowledge graphs, controlled audits
Languages	Mandarin Chinese (native); English (IELTS 6.5)